# Functions of Several Variables & Partial Derivatives

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Verify Clairaut's Theorem on equality of mixed partial derivatives for the function $f(x, y) = \cos(2x - 3y) + x^2y^3$ by computing $f_{xy}$ and $f_{yx}$ directly.

1. Compute $f_x$ and $f_{xy}$: $f_x = \frac{\partial}{\partial x}[\cos(2x - 3y) + x^2y^3] = -2\sin(2x - 3y) + 2xy^3$. $f_{xy} = \frac{\partial}{\partial y}[-2\sin(2x - 3y) + 2xy^3] = -2\cos(2x - 3y)(-3) + 6xy^2 = 6\cos(2x - 3y) + 6xy^2$. 2. Compute $f_y$ and $f_{yx}$: $f_y = \frac{\partial}{\partial y}[\cos(2x - 3y) + x^2y^3] = 3\sin(2x - 3y) + 3x^2y^2$. $f_{yx} = \frac{\partial}{\partial x}[3\sin(2x - 3y) + 3x^2y^2] = 3\cos(2x - 3y)(2) + 6xy^2 = 6\cos(2x - 3y) + 6xy^2$. Since $f$ is $C^2$ on $\mathbb{R}^2$.

**2** **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Let $f(x, y) = x^3e^{2y} - y\ln(x)$ defined on $x > 0$. Compute the first-order partial derivatives $f_x(x, y)$ and $f_y(x, y)$, and evaluate $f_x(1, 0)$ and $f_y(1, 0)$.

1. Partial Derivative with respect to $x$ (treating $y$ as constant): $f_x(x, y) = \frac{\partial}{\partial x}[x^3e^{2y}] - \frac{\partial}{\partial x}[y\ln x] = 3x^2e^{2y} - \frac{y}{x}$. At $(1, 0)$: $f_x(1, 0) = 3(1)^2e^0 - \frac{0}{1} = 3$. 2. Partial Derivative with respect to $y$ (treating $x$ as constant): $f_y(x, y) = \frac{\partial}{\partial y}[x^3e^{2y}] - \frac{\partial}{\partial y}[y\ln x] = 2x^3e^{2y} - \ln(x)$. At $(1, 0)$: $f_y(1, 0) = 2(1)^3e^0 - \ln(1) = 2 - 0 = 2$.

**3** **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A temperature distribution is given by $T(x, y, z) = \frac{120}{1 + x^2 + 2y^2 + 3z^2}$. At the point $P(1, 1, 1)$, find: (a) the direction of steepest temperature increase, (b) the maximum rate of increase, and (c) the equation of the tangent plane to the isothermal level surface passing through $P$.

1. Compute the gradient $ablaT$: Let $u = 1 + x^2 + 2y^2 + 3z^2$. At $P(1, 1, 1)$, $u = 1 + 1 + 2 + 3 = 7$. $ablaT = -\frac{120}{u^2}\langle 2x, 4y, 6z\rangle$. At $P(1, 1, 1)$: $ablaT(1, 1, 1) = -\frac{120}{49}\langle 2, 4, 6\rangle = -\frac{240}{49}\langle 1, 2, 3\rangle$. (a) Direction of steepest increase: $\mathbf{u}_{\max} = \frac{ablaT}{||ablaT||} = -\frac{\langle 1, 2, 3\rangle}{\sqrt{1 + 4 + 9}} = -\frac{1}{\sqrt{14}}\langle 1, 2, 3\rangle$. (b) Maximum rate of increase.

**4** **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Evaluate the double integral of the function $f(x, y) = 2xy$ over the rectangular region $R = [0, 2] \times [0, 3]$ using Fubini's Theorem. Set up the iterated integral by integrating with respect to $y$ first, then $x$.

To evaluate the double integral $\iint_R 2xy\,dA$ where $R = [0, 2] \times [0, 3]$, we can use Fubini's Theorem. We will set up the iterated integral by integrating with respect to $y$ first, then $x$: $\int_0^2 \int_0^3 2xy\,dy\,dx$ First, integrate with respect to $y$, treating $x$ as a constant: $\int_0^3 2xy\,dy = \left[xy^2\right]_{y=0}^{y=3} = x(3^2) - x(0^2) = 9x$ Now, integrate the result with respect to $x$.

© ClipSAT — clipsat.org | All Rights Reserved

**5** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the vector field $\mathbf{F}(x, y) = \langle x^2, y \rangle$. According to Green's Theorem, the line integral of $\mathbf{F}$ around a simple closed curve $C$ is equal to the double integral of $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$ over the region $D$ enclosed by $C$, where $\mathbf{F} = \langle P, Q \rangle$. For the given vector field $\mathbf{F}(x, y) = \langle x^2, y \rangle$, identify the value of $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$.

The given vector field is $\mathbf{F}(x, y) = \langle P(x, y), Q(x, y) \rangle = \langle x^2, y \rangle$. According to Green's Theorem, we need to compute $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$. First, find the partial derivative of $Q$ with respect to $x$: $\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(y) = 0$. Next, find the partial derivative of $P$ with respect to $y$: $\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(x^2) = 0$. Therefore.

**6** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Let $z = f(x, y)$ be a differentiable function, with polar coordinates transformation $x = r\cos\theta$ and $y = r\sin\theta$. Express $\frac{\partial z}{\partial r}$ and $\frac{\partial z}{\partial \theta}$ in terms of $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$, and prove that: $\left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2 = \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = ||ablaz||^2$.

1. Apply the Multivariable Chain Rule: $\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial r} = z_x\cos\theta + z_y\sin\theta$.
$\frac{\partial z}{\partial \theta} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial \theta} = z_x(-r\sin\theta) + z_y(r\cos\theta) = r(-z_x\sin\theta + z_y\cos\theta)$. 2. Divide by $r$ and square both expression.

**7** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Use the method of Lagrange multipliers to find the maximum volume of a rectangular box with faces parallel to the coordinate planes inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

Let the corner of the box in the first octant be $(x, y, z)$ with $x, y, z > 0$. The total volume is $V(x, y, z) = (2x)(2y)(2z) = 8xyz$. The constraint is $g(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 = 0$. 1. Set up the Lagrange multiplier system $ablaV = \lambda ablag$: $8yz = \lambda\frac{2x}{a^2}, \quad 8xz = \lambda\frac{2y}{b^2}, \quad 8xy = \lambda\frac{2z}{c^2}$. 2. Multiply the three equations by $x, y, z$ respectively.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Derive the Gaussian integral $I = \int_{-\infty}^{\infty} e^{-x^2/2} dx = \sqrt{2\pi}$ from first principles by evaluating the double integral $I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)/2} dA$ in polar coordinates.

1. Express $I^2$ as a double integral: By Fubini's theorem on independent variables:
$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2/2}\,dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2/2}\,dy\right) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)/2}\,dx\,dy$. 2. Convert to polar coordinates: Let $x = r\cos\theta, y = r\sin\theta$, so $x^2 + y^2 = r^2$ and $dA = r\,dr\,d\theta$. The domain $\mathbb{R}^2$ maps to $0 \le r < \infty$ and $0 \le \theta \le 2\pi$.

© ClipSAT — clipsat.org | All Rights Reserved

**9** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the region $R$ in the first quadrant bounded by the x-axis, the y-axis, and the line $y = -2x + 6$. We are interested in evaluating the double integral $\iint_R f(x, y)\,dA$ where $f(x, y) = x + y$. However, the integral is set up in polar coordinates as $\int_{\alpha}^{\beta}\int_{r_1(\theta)}^{r_2(\theta)} f(r\cos\theta, r\sin\theta)\cdot r\,dr\,d\theta$. If this integral is evaluated from $\theta = 0$ to $\theta = \frac{\pi}{2}$, and the inner integral bounds are $r_1(\theta) = 0$ and $r_2(\theta) = \frac{6}{\sin\theta + 2\cos\theta}$, and the outer integral is $\int_0^{\pi/2}\int_0^{\frac{6}{\sin\theta+2\cos\theta}} (r\cos\theta + r\sin\theta)\cdot r\,dr\,d\theta$, determine the value of $\alpha$ and $\beta$ that correspond to the description of region $R$ in polar coordinates for $0 \le \theta \le \frac{\pi}{2}$.

The region R is in the first quadrant, bounded by the x-axis ($y = 0$), the y-axis ($x = 0$), and the line $y = -2x + 6$. When $x = 0$, $y = 6$. When $y = 0$, $2x = 6$, so $x = 3$. The vertices of the region are $(0, 0), (3, 0), (0, 6)$. In polar coordinates, $x = r\cos\theta$ and $y = r\sin\theta$. The line $y = -2x + 6$ becomes $r\sin\theta = -2r\cos\theta + 6$. Rearranging for $r$, we get $r(\sin\theta + 2\cos\theta) = 6$, so $r = \frac{6}{\sin\theta + 2\cos\theta}$. The first quadrant is defined by.

**10** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate the non-elementary integral $\int_0^1\int_y^1 e^{x^2}\,dx\,dy$ by reversing the order of integration.

1. Sketch the region of integration $D$: The given bounds are $y \le x \le 1$ for $0 \le y \le 1$. This describes the triangle bounded by the lines $y = 0$, $x = 1$, and $y = x$. 2. Reverse the order to $dy\,dx$: In terms of vertical cross-sections, $x$ ranges from $0$ to $1$, and for each fixed $x$, $y$ ranges from $0$ up to the line $y = x$: $\int_0^1\int_0^x e^{x^2}\,dy\,dx$. 3. Evaluate the inner integral: $\int_0^x e^{x^2}\,dy = e^{x^2}[y]_0^x = xe^{x^2}$. 4. Evaluate the outer integral: Substitute.

**11** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Use Stokes' Theorem to evaluate $\oint_C \mathbf{F}\cdot d\mathbf{r}$, where $\mathbf{F} = \langle -y^3, x^3, -z^3\rangle$ and $C$ is the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = 1$, oriented counterclockwise when viewed from above. Verify surface independence by comparing the flux of $abla \times \mathbf{F}$ through the flat disk $S_1 : x^2 + y^2 \le 1, z = 1$ and the paraboloid cap $S_2 : z = 2 - (x^2 + y^2)$.

1. Compute $abla \times \mathbf{F}$: $abla \times \mathbf{F} = \left|\mathbf{i}, \mathbf{j}, \mathbf{k}; \partial_x, \partial_y, \partial_z; -y^3, x^3, -z^3\right| = \langle 0 - 0,\ 0 - 0,\ 3x^2 - (-3y^2)\rangle = \langle 0,\ 0,\ 3(x^2 + y^2)\rangle$. 2. Flux through flat disk $S_1$ ($z = 1$, $\mathbf{n} = \mathbf{k}$): On $S_1$, $d\mathbf{S} = \mathbf{k}\,dA$. $\iint_{S_1} (abla \times \mathbf{F})\cdot d\mathbf{S} = \iint_D 3(x^2 + y^2)\,dA$. In polar coordinates.

© ClipSAT — clipsat.org | All Rights Reserved

**12** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

A particle's position in three-dimensional space is given by the vector function $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$, where $t$ represents time. The velocity of the particle is given by $\mathbf{v}(t) = \mathbf{r}'(t)$. Suppose the velocity components are related by the partial differential equations: $\frac{\partial v_x}{\partial t} = -2x(t) + 3y(t)$ and $\frac{\partial v_y}{\partial t} = x(t) - 4y(t)$. If at time $t = 0$, the particle is at the origin $(0, 0, 0)$ and its velocity is $\mathbf{v}(0) = \langle 5, -2, 0 \rangle$, determine the value of $\frac{\partial^2 v_x}{\partial t^2}$ at $t = 0$.

We are given the partial differential equations relating the velocity components: $\frac{\partial v_x}{\partial t} = -2x(t) + 3y(t)$ and $\frac{\partial v_y}{\partial t} = x(t) - 4y(t)$. To find $\frac{\partial^2 v_x}{\partial t^2}$, we differentiate the first equation with respect to $t$: $\frac{\partial^2 v_x}{\partial t^2} = \frac{\partial}{\partial t}(-2x(t) + 3y(t))$. Using the chain rule, this becomes $\frac{\partial^2 v_x}{\partial t^2} = -2\frac{\partial x}{\partial t} + 3\frac{\partial y}{\partial t}$. Sin.

**13** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Use the Divergence Theorem to calculate the outward flux of $\mathbf{F} = \langle x^3 + y\sin z,\ y^3 + z\cos x,\ z^3 + xy \rangle$ across the closed boundary surface of the solid cylinder $E : x^2 + y^2 \le a^2,\ 0 \le z \le h$.

1. Compute the divergence $abla \cdot \mathbf{F}$: $abla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x^3 + y\sin z) + \frac{\partial}{\partial y}(y^3 + z\cos x) + \frac{\partial}{\partial z}(z^3 + xy) = 3x^2 + 3y^2 + 3z^2 = 3(x^2 + y^2 + z^2)$. Notice that all cross-coupling transcendental terms vanish identically. 2. Apply the Divergence Theorem: $\iint_{\partial E} \mathbf{F} \cdot d\mathbf{S} = \iiint_E 3(x^2 + y^2 + z^2)\, dV$. 3. Convert to cylindrical coordinates: In $E$, $x^2 + y^2 = r^2$, $0 \le r \le a$.

**14** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Use the Green's Theorem line integral formula for area, $A = \frac{1}{2}\oint_C (x\,dy - y\,dx)$, to derive the area enclosed by the hypocycloid (astroid) $\mathbf{r}(t) = \langle a\cos^3 t,\ a\sin^3 t \rangle$ for $0 \le t \le 2\pi$.

1. Compute the differentials: $dx = -3a\cos^2 t\sin t\,dt$, $dy = 3a\sin^2 t\cos t\,dt$. 2. Evaluate $x\,dy - y\,dx$: $x\,dy = (a\cos^3 t)(3a\sin^2 t\cos t\,dt) = 3a^2\cos^4 t\sin^2 t\,dt$. $y\,dx = (a\sin^3 t)(-3a\cos^2 t\sin t\,dt) = -3a^2\sin^4 t\cos^2 t\,dt$. $x\,dy - y\,dx = 3a^2\cos^2 t\sin^2 t(\cos^2 t + \sin^2 t)\,dt = 3a^2\cos^2 t\sin^2 t\,dt$. 3. Use trigonometric identities: Since $\cos t\sin t = \frac{1}{2}\sin(2t)$.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 4.01 Double Integrals**

Let the function $f(x, y)$ be defined as $f(x, y) = 2x^2 - 3y^2$. Calculate the partial derivative of $f$ with respect to $y$ at the point $(2, 3)$.

The function is given by $f(x, y) = 2x^2 - 3y^2$. To find the partial derivative of $f$ with respect to $y$, denoted as $f_y$, we treat $x$ as a constant and differentiate with respect to $y$. $f_y(x, y) = \frac{\partial}{\partial y}(2x^2 - 3y^2)$ Differentiating the term $2x^2$ with respect to $y$ yields 0, as it is a constant with respect to $y$. Differentiating the term $-3y^2$ with respect to $y$ yields $-3 \cdot 2y = -6y$. Therefore, the partial derivative of $f$ with respect to $y$ is: $f_y(x, y) = -6y$ Now we n.

© ClipSAT — clipsat.org | All Rights Reserved

**16** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Consider the vector field $\mathbf{F}(x, y, z) = \langle x, y, z \rangle$. Let $S$ be the closed surface of the unit cube defined by $0 \le x \le 1, 0 \le y \le 1, 0 \le z \le 1$, oriented outwards. According to the Divergence Theorem, the flux of $\mathbf{F}$ through $S$ is equal to the triple integral of the divergence of $\mathbf{F}$ over the volume $V$ enclosed by $S$. Calculate this flux by evaluating $\iiint_V abla \cdot \mathbf{F}\, dV$.

The vector field is $\mathbf{F}(x, y, z) = \langle x, y, z \rangle$. The divergence of $\mathbf{F}$ is $abla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1 = 3$. The volume $V$ enclosed by the unit cube is given by $0 \le x \le 1, 0 \le y \le 1, 0 \le z \le 1$. The volume of this cube is $1 \times 1 \times 1 = 1$. Therefore, the flux is given by the triple integral: $\iiint_V abla \cdot \mathbf{F}\, dV = \iiint_V 3\, dV$. Since.

**17** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Use Green's Theorem to evaluate $\oint_C (y^2 - 7y)\, dx + (2xy + 2x)\, dy$, where $C$ is the positively oriented boundary of the region $D$ enclosed between the parabolas $y = x^2$ and $x = y^2$.

1. Compute the 2D curl: With $P = y^2 - 7y$ and $Q = 2xy + 2x$: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = (2y + 2) - (2y - 7) = 2 - (-7) = 9$. 2. Intersections of boundary curves: $y = x^2$ and $x = y^2 \Rightarrow x = x^4 \Rightarrow x = 0, 1$. Region $D$ is described by $0 \le x \le 1$ and $x^2 \le y \le \sqrt{x}$. 3. Apply Green's Theorem.

**18** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Double Integrals

Use Green's Theorem to evaluate the circulation $\oint_C (3y - e^{\sin x})\, dx + (7x + \sqrt{y^4 + 1})\, dy$, where $C$ is the circle $x^2 + y^2 = 9$ oriented counterclockwise.

1. Identify components and compute 2D curl: With $P = 3y - e^{\sin x}$ and $Q = 7x + \sqrt{y^4 + 1}$: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \frac{\partial}{\partial x}\left(7x + \sqrt{y^4 + 1}\right) - \frac{\partial}{\partial y}(3y - e^{\sin x}) = 7 - 3 = 4$. The non-elementary transcendental terms vanish completely. 2. Apply Green's Theorem: The region $D$ is the circular disk of radius $R = 3$.

**19** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Consider the function $f(x, y) = 3x^2 - 5y^3$. Find the partial derivative of $f$ with respect to $y$ evaluated at the point $(2, -1)$, i.e., $f_y(2, -1)$.

To find the partial derivative of $f(x, y) = 3x^2 - 5y^3$ with respect to $y$, we treat $x$ as a constant and differentiate with respect to $y$. $f_y(x, y) = \frac{\partial}{\partial y}(3x^2 - 5y^3)$ Since $3x^2$ is a constant with respect to $y$, its derivative is 0. The derivative of $-5y^3$ with respect to $y$ is $-5 \cdot 3y^{3-1} = -15y^2$. Therefore, $f_y(x, y) = -15y^2$. Now, we evaluate $f_y(x, y)$ at the point $(2, -1)$. $f_y(2, -1) = -15(-1)^2$ $f_y(2, -1) = -15(1)$ $f_y(2, -1) = -15$.

© ClipSAT — clipsat.org | All Rights Reserved

**20** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Find the flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$ of $\mathbf{F} = \langle x, y, z \rangle$ across the part of the plane $x + y + z = 3$ in the first octant, with upward normal.

Write $z = -x - y + 3$; $d\mathbf{S} = \langle 1, 1, 1 \rangle \, dA$. The flux is $\iint_D \mathbf{F} \cdot \langle 1, 1, 1 \rangle \, dA = \frac{27}{2}$. Final Answer: $\frac{27}{2}$ Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

---

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the domain of the function $f(x, y) = \ln(9 - x^2 - y^2)$.

**a.** **The interior of the circle** $x^2 + y^2 < 9$ ✓

**b.** The exterior of the circle $x^2 + y^2 > 9$

**c.** The closed disk $x^2 + y^2 \leq 9$

**d.** All of $\mathbb{R}^2$

The argument of the natural logarithm must be strictly positive: $9 - x^2 - y^2 > 0 \Leftrightarrow x^2 + y^2 < 9$, which is the open interior of a circle of radius 3 centered at the origin.

**22** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What geometric curves are the level curves of $z = x^2 + y^2$ for $c > 0$?

**a.** **Concentric circles centered at** $(0, 0)$ ✓

**b.** Lines of slope 1

**c.** Parabolas opening upward

**d.** Hyperbolas

Setting $z = c \Rightarrow x^2 + y^2 = c$. For $c > 0$, this is a family of concentric circles of radius $\sqrt{c}$.

© ClipSAT — clipsat.org | All Rights Reserved

**23** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)
What is the range of $g(x, y) = \sqrt{16 - x^2 - y^2}$?

**a.** $[0, 4]$ ✓

**b.** $[0, 16]$

**c.** $[-4, 4]$

**d.** $[0, \infty)$

When $x^2 + y^2 = 0$, $g(0, 0) = \sqrt{16} = 4$. On the boundary $x^2 + y^2 = 16$, $g = 0$. Since square roots are non-negative, the range is $[0, 4]$.

**24** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)
Find the partial derivative $f_x(x, y)$ for $f(x, y) = 3x^3y^2 - 5x^2 + 4y^3$.

**a.** $9x^2y^2 - 10x$ ✓

**b.** $6x^3y + 12y^2$

**c.** $9x^2y^2 - 10x + 12y^2$

**d.** $3x^2y^2 - 10x$

Treat $y$ as a constant: $\frac{\partial}{\partial x}[3x^3y^2] = 3(3x^2)y^2 = 9x^2y^2$; $\frac{\partial}{\partial x}[-5x^2] = -10x$; $\frac{\partial}{\partial x}[4y^3] = 0$. Result: $9x^2y^2 - 10x$.

**25** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)
What does Clairaut's Theorem state about mixed second-order partial derivatives $f_{xy}$ and $f_{yx}$?

**a.** **If both $f_{xy}$ and $f_{yx}$ are continuous on an open region, then $f_{xy} = f_{yx}$** ✓

**b.** Mixed partials are always equal to 0

**c.** The sum $f_{xy} + f_{yx} = 0$

**d.** Equality holds only for polynomials

Clairaut's Theorem guarantees that mixed partial derivatives are equal ($f_{xy} = f_{yx}$) provided they are continuous on an open disk.

**26** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)
For $f(x, y) = x^3y^4$, verify Clairaut's Theorem by finding $f_{xy}$.

**a.** $12x^2y^3$ ✓

**b.** $3x^2y^4$

**c.** $4x^3y^3$

**d.** $6xy^2$

$f_x = 3x^2y^4$. Then $f_{xy} = \frac{\partial}{\partial y}[3x^2y^4] = 3x^2(4y^3) = 12x^2y^3$.

© ClipSAT — clipsat.org | All Rights Reserved

**27** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the equation of the tangent plane to the surface $z = f(x, y)$ at the point $(x_0, y_0, z_0)$?

**a.** $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$ ✓

**b.** $z - z_0 = f_x(x - x_0) - f_y(y - y_0)$

**c.** $z = f_x x_0 + f_y y_0$

**d.** $(x - x_0) + (y - y_0) + (z - z_0) = 0$

The linear approximation (tangent plane) formula is $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$.

**28** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the equation of the tangent plane to $z = x^2 + y^2$ at $(1, 2, 5)$.

**a.** $z - 5 = 2(x - 1) + 4(y - 2) \Leftrightarrow 2x + 4y - z = 5$ ✓

**b.** $2x + 4y + z = 15$

**c.** $x + 2y - z = 0$

**d.** $4x + 2y - z = 5$

$f_x = 2x \Rightarrow f_x(1, 2) = 2$. $f_y = 2y \Rightarrow f_y(1, 2) = 4$. Plane: $z - 5 = 2(x - 1) + 4(y - 2) \Rightarrow z - 5 = 2x - 2 + 4y - 8 \Rightarrow 2x + 4y - z = 5$.

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is a normal vector to the surface $z = f(x, y)$ at $(x_0, y_0)$?

**a.** $\mathbf{n} = [f_x(x_0, y_0), f_y(x_0, y_0), -1]$ ✓

**b.** $\mathbf{n} = [f_x, f_y, 1]$

**c.** $\mathbf{n} = [1, 1, 1]$

**d.** $\mathbf{n} = [-f_y, f_x, 0]$

Rewriting as $F(x, y, z) = f(x, y) - z = 0$, the gradient $\nabla F = [f_x, f_y, -1]$ forms an outward normal vector.

© ClipSAT — clipsat.org | All Rights Reserved

**30** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the function $f(x, y) = 5x^3 - 2y^2 + 7$. Compute the value of the partial derivative $f_y(2, -3)$.

**a.** 20

**b.** **-12 ✓**

**c.** 12

**d.** -20

To find the partial derivative of $f(x, y)$ with respect to $y$, we treat $x$ as a constant. Given $f(x, y) = 5x^3 - 2y^2 + 7$. The partial derivative with respect to $y$ is denoted by $f_y$. $f_y = \frac{\partial}{\partial y}(5x^3 - 2y^2 + 7)$ Applying the power rule and noting that $5x^3$ and $7$ are constants with respect to $y$: $f_y = 0 - 2(2y) + 0$ $f_y = -4y$ Now, we need to evaluate $f_y$ at the point $(2, -3)$: $f_y(2, -3) = -4(-3)$ $f_y(2, -3) = 12$

**31** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Evaluate the double integral $\iint_R xy\,dA$ where $R$ is the region in the first quadrant bounded by the lines $y = x$, $x = 1$, and $y = 0$.

**a.** 1/12

**b.** **1/8 ✓**

**c.** 1/6

**d.** 1/4

The region of integration $R$ is a triangle with vertices (0,0), (1,0), and (1,1). We can set up the double integral using Fubini's Theorem. Since $y = x$ is the upper bound for $y$ and $x = 1$ is the upper bound for $x$, we can integrate with respect to $y$ first, then $x$. The limits for $y$ are from 0 to $x$, and the limits for $x$ are from 0 to 1. $\iint_R xy\,dA = \int_0^1 \int_0^x xy\,dy\,dx$ First, evaluate the inner integral with respect to $y$.

**32** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the region $R$ in the first quadrant bounded by the x-axis, the line $y = x$, and the circle $x^2 + y^2 = 16$. The integral $\iint_R f(x, y)\,dA$ can be evaluated more easily in polar coordinates. If $f(x, y) = \frac{1}{\sqrt{x^2 + y^2}}$, and the integral in polar coordinates is given by $\int_0^{\pi/4} \int_0^4 r \cdot \frac{1}{r}\,dr\,d\theta = \int_0^{\pi/4} A\,d\theta$, find the value of $A$.

**a.** **4 ✓**

**b.** $\pi/4$

**c.** 16

**d.** 2

The region R is described in the first quadrant, bounded by the x-axis ($\theta = 0$), the line $y = x$ ($\theta = \pi/4$), and the circle $x^2 + y^2 = 16$ ($r = 4$). The integrand is $f(x, y) = \frac{1}{\sqrt{x^2 + y^2}}$. In polar coordinates, $x^2 + y^2 = r^2$, so $\sqrt{x^2 + y^2} = r$. Thus, $f(x, y) = \frac{1}{r}$. The differential area element in polar coordinates is $dA = r\,dr\,d\theta$. The double integral in polar coordinates is.

© ClipSAT — clipsat.org | All Rights Reserved

**33** [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For $f(x, y) = 4x^2y^2 + 3xy$, find the mixed partial $f_{xy}(1, -2)$.

**a.** -32

**b.** 26

**c.** 29

**d.** **-29 ✓**

$f_x = 8xy^2 + 3y$, so $f_{xy} = \frac{\partial}{\partial y} f_x = 16xy + 3$. At $(1, -2)$: $f_{xy} = -29$ (matches $f_{yx}$, as Clairaut's theorem guarantees since $f$ is a polynomial).

**34** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $z = f(x, y) = x^2 + y^2$. Let $x = 2t$ and $y = 3t$. Use the Multivariable Chain Rule to find $\frac{dz}{dt}$ when $t = 1$.

**a.** 10

**b.** 13

**c.** **26 ✓**

**d.** 52

We are given $z = f(x, y) = x^2 + y^2$, with $x = 2t$ and $y = 3t$. We need to find $\frac{dz}{dt}$ using the Multivariable Chain Rule: $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$. First, find the partial derivatives of $z$ with respect to $x$ and $y$: $\frac{\partial z}{\partial x} = 2x$ $\frac{\partial z}{\partial y} = 2y$ Next, find the derivatives of $x$ and $y$ with respect to $t$: $\frac{dx}{dt} = 2$ $\frac{dy}{dt} = 3$ Now, substitute these into.

**35** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the region $R$ in the $xy$-plane defined by $0 \le x \le 1$ and $0 \le y \le x^2$. Evaluate the double integral $\iint_R y\,dA$.

**a.** $\frac{1}{12}$ **✓**

**b.** $\frac{1}{10}$

**c.** $\frac{1}{8}$

**d.** $\frac{1}{6}$

The double integral can be evaluated using Fubini's Theorem as an iterated integral. The region of integration is described by $0 \le x \le 1$ and $0 \le y \le x^2$. Therefore, the integral is set up as: $\int_0^1 \int_0^{x^2} y\,dy\,dx$ First, we integrate with respect to $y$: $\int_0^{x^2} y\,dy = \left[\frac{y^2}{2}\right]_0^{x^2} = \frac{(x^2)^2}{2} - \frac{0^2}{2} = \frac{x^4}{2}$ Now, we integrate the result with respect to $x$.

© ClipSAT — clipsat.org | All Rights Reserved

**36** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let the scalar field be given by $f(x, y) = 3x^2 - 5y$. Compute the value of $f_x(2, -3)$.

**a.** 6

**b.** **12** ✓

**c.** -6

**d.** 15

To find $f_x(2, -3)$, we first compute the partial derivative of $f(x, y)$ with respect to $x$. Treating $y$ as a constant, we get: $f_x(x, y) = \frac{\partial}{\partial x}(3x^2 - 5y) = 6x$. Now, we evaluate $f_x$ at the point $(2, -3)$: $f_x(2, -3) = 6(2) = 12$.

**37** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Under the change of variables $x = 4u - 2v$, $y = -u + 4v$, a region $S$ in the $uv$-plane with area 3 maps to a region $R$. Find the area of $R$.

**a.** 3

**b.** 14

**c.** 42 ✓

**d.** 24

$\left|\frac{\partial(x, y)}{\partial(u, v)}\right| = |14|$, so $\text{Area}(R) = |J| \cdot \text{Area}(S) = 42$.

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Double Integrals

Euler's theorem for homogeneous functions: if $f(tx, ty) = t^n f(x, y)$, then $xf_x + yf_y =$

**a.** $f$

**b.** $nf$ ✓

**c.** $n(n-1)f$

**d.** 0

Applying Partial Derivatives and Gradient principles, Euler: $xf_x + yf_y = nf$. Therefore, the verified correct option is $nf$.

© ClipSAT — clipsat.org | All Rights Reserved

**39** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Gradient descent step: $\mathbf{x}_{n+1} = \mathbf{x}_n - \alpha ablaf(\mathbf{x}_n)$. For $f = x^2 + y^2$ at $(1, 1)$, $\alpha = 0.5$:

**a.** $(0.5, 0.5)$

**b.** $(0, 0)$ ✓

**c.** $(-1, -1)$

**d.** $(1.5, 1.5)$

$ablaf = (2, 2)$; step: $(1, 1) - 0.5(2, 2) = (0, 0)$.

**40** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Consider the vector field $\mathbf{F}(x, y) = \langle 2x, y \rangle$. Evaluate the line integral $\oint_C \mathbf{F} \cdot d\mathbf{r}$ where $C$ is the circle $x^2 + y^2 = 4$ oriented counterclockwise. Use Green's Theorem.

**a.** $0$

**b.** $8\pi$ ✓

**c.** $4\pi$

**d.** $16\pi$

According to Green's Theorem, $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$, where $\mathbf{F} = \langle P, Q \rangle$ and $D$ is the region enclosed by $C$. ; Here, $P(x, y) = 2x$ and $Q(x, y) = y$. ; We compute the partial derivatives: $\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(y) = 0$ and $\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(2x) = 0$. ; Therefore.

**41** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Double Integrals

Consider the function $f(x, y) = x^2 + y^2 - 4x - 6y + 13$. Find the coordinates of the critical point for this function.

**a.** **(2, 3)** ✓

**b.** (-2, -3)

**c.** (4, 6)

**d.** (0, 0)

To find the critical point(s) of a function of two variables, we need to find where the gradient is zero. The gradient is given by $ablaf = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$. First, we compute the partial derivative with respect to x: $\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(x^2 + y^2 - 4x - 6y + 13) = 2x - 4$. Next, we compute the partial derivative with respect to y.

© ClipSAT — clipsat.org | All Rights Reserved

**42** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Consider the solid region $E$ in the first octant bounded by the coordinate planes and the surface $z = 4 - x^2 - y^2$. The volume of this region is given by the triple integral $V = \iiint_E dV$. We are given that the volume can also be expressed in cylindrical coordinates as $V = \int_0^{\pi/2} \int_0^{\sqrt{a}} \int_0^{4-r^2} r\,dz\,dr\,d\theta$. Find the value of the parameter $a$.

**a.** 2

**b.** **4** ✓

**c.** 16

**d.** 8

The solid region $E$ is in the first octant and is bounded by $z = 0$, $x = 0$, $y = 0$ and $z = 4 - x^2 - y^2$. In cylindrical coordinates, $x^2 + y^2 = r^2$, so the surface is $z = 4 - r^2$. Since the region is in the first octant, $0 \le \theta \le \pi/2$. The lower bound for $z$ is $z = 0$. The upper bound for $z$ is $4 - r^2$. \\ To determine the bounds for $r$, we look at the projection of the solid onto the $xy$-plane. This projection is determined by the intersection of $z = 4 - x^2 - y^2$ with the $xy$-p.

**43** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Consider the solid region $E$ bounded below by the cone $z = \sqrt{x^2 + y^2}$ and above by the sphere $x^2 + y^2 + z^2 = 16$. The volume of $E$ is given by the triple integral $V = \iiint_E dV$. If $E$ is expressed in spherical coordinates, the integral becomes $\int_\alpha^\beta \int_\gamma^\delta \int_{\rho_1}^{\rho_2} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$. Determine the value of $\alpha$, the lower limit of integration for $\theta$.

**a.** **0** ✓

**b.** $\frac{\pi}{2}$

**c.** $\pi$

**d.** $2\pi$

The region $E$ is a spherical cone. The upper bound is a sphere of radius 4 centered at the origin, so $\rho$ ranges from 0 to 4. The lower bound is the cone $z = \sqrt{x^2 + y^2}$. In spherical coordinates, $z = \rho\cos\phi$ and $\sqrt{x^2 + y^2} = \rho\sin\phi$. Thus, the cone equation becomes $\rho\cos\phi = \rho\sin\phi$. For $\rho eq 0$, this simplifies to $\cos\phi = \sin\phi$, which means $\tan\phi = 1$. In the range $0 \le \phi \le \pi$, this occurs at $\phi = \frac{\pi}{4}$. Therefore.

© ClipSAT — clipsat.org | All Rights Reserved

**44** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Double Integrals

Consider the function $f(x, y) = x^3 + y^3 - 3axy$, where $a$ is a positive constant. It is known that $f(x, y)$ has a critical point at $(x_0, y_0)$ where $x_0 = 2$ and $y_0 = 2$. Find the value of $a$.

**a.** 1

**b.** **2 ✓**

**c.** 3

**d.** 4

To find the critical points of the function $f(x, y) = x^3 + y^3 - 3axy$, we first compute the first partial derivatives and set them to zero: $f_x = 3x^2 - 3ay$ $f_y = 3y^2 - 3ax$ Setting these to zero, we get the system of equations: 1) $3x^2 - 3ay = 0$ => $x^2 = ay$ 2) $3y^2 - 3ax = 0$ => $y^2 = ax$ We are given that $(x_0, y_0) = (2, 2)$ is a critical point. Substituting these values into the equations: From equation 1: $2^2 = a(2)$ => $4 = 2a$ => $a = 2$. From equation 2: $2^2 = a(2)$ => $4 = 2a$ => $a = 2$. Bo.

**45** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Consider the solid region $E$ defined by $0 \le x \le 1$, $0 \le y \le 2$, and $0 \le z \le 3$. Evaluate the triple integral of the function $f(x, y, z) = 2x + 3y$ over the region $E$.

**a.** **18 ✓**

**b.** 15

**c.** 21

**d.** 24

The triple integral of $f(x, y, z) = 2x + 3y$ over the rectangular box $E$ is given by: $\iiint_E (2x + 3y)\,dV = \int_0^3 \int_0^2 \int_0^1 (2x + 3y)\,dx\,dy\,dz$ First, integrate with respect to $x$: $\int_0^1 (2x + 3y)\,dx = [x^2 + 3yx]_0^1 = (1^2 + 3y(1)) - (0^2 + 3y(0)) = 1 + 3y$ Next, integrate with respect to $y$: $\int_0^2 (1 + 3y)\,dy = \left[y + \frac{3}{2}y^2\right]_0^2 = (2 + \frac{3}{2}(2^2)) - (0 + \frac{3}{2}(0)^2) = 2 + \frac{3}{2}(4) = 2 + 6 = 8$.

© ClipSAT — clipsat.org | All Rights Reserved